

DATA SCIENCE NOTES

1. Introduction to Data Science

Data Science is the process of collecting, analyzing, and interpreting data to extract useful insights.

2. Key Components

- Data Collection
- Data Cleaning
- Data Analysis
- Data Visualization
- Decision Making

3. Tools & Technologies

Python, R, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

4. Types of Data

- Structured Data
- Unstructured Data
- Semi-structured Data

DATA SCIENCE NOTES (Continued)

5. Machine Learning Basics

Types:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

6. Applications

Healthcare, Finance, E-commerce, Social Media

7. Advantages

- Better decision making
- Automation
- Improved accuracy

8. Challenges

- Data privacy
- Big data handling
- Skilled professionals required

9. Conclusion

Data Science helps convert raw data into meaningful insights for better decisions.